

One Rating, Multiple Services

Solving the 100-Service Dilemma: Automated Attribution for Multi-Service Public Satisfaction Surveys

Abdul Aziz Nurussadad

Badan Informasi Geospasial of Republic of Indonesia

R · tidyverse · Public Sector Data



THE PROBLEM

Nine components. Hundreds of services. One exhausted respondent.

2

PERMENPAN RB 14/2017

9

required quality components per service
(procedures, cost, staff, facilities ...)

Mandatory for every public service in Indonesia

A ONE-STOP SERVICE MALL

100

 services × 9 components

= 900 answers from a single visitor

Respondent fatigue → abandoned surveys →
compromised data quality



THE SHIFT

From one form per service, to one holistic rating per visit

NAIVE APPROACH

- One survey per service used
- Technically clean data
- Practically unusable — nobody
- finishes a 100-part form



OUR APPROACH

- Check off every service used this visit
- One shared rating across the 9 components
- Matches how people actually
- experience a visit



THE R WORKFLOW

Disaggregate, then attribute

1

Google Form export

Multi-select answers
arrive as one
semicolon string:
"Service A; Service B;
Service E"

2

Disaggregate

`separate_longer_delim()`
explodes
one row into three —
one per
service, same ratings
copied to each

3

Attribute

`group_by(service) +
summarise()`
averages every rating
that
touched that service



THE CODE

Four lines from raw export to compliant score

5

```
clean_data <- raw_data %>%  
  separate_longer_delim(services, delim = "; ") %>%      # 1. disaggregate  
  group_by(services) %>%  
  summarise(across(U1:U9, mean), n = n()) %>%              # 2. attribute  
  mutate(  
    IKM = rowSums(across(starts_with("U")) * 0.111) * 25 # 3. score  
  )
```

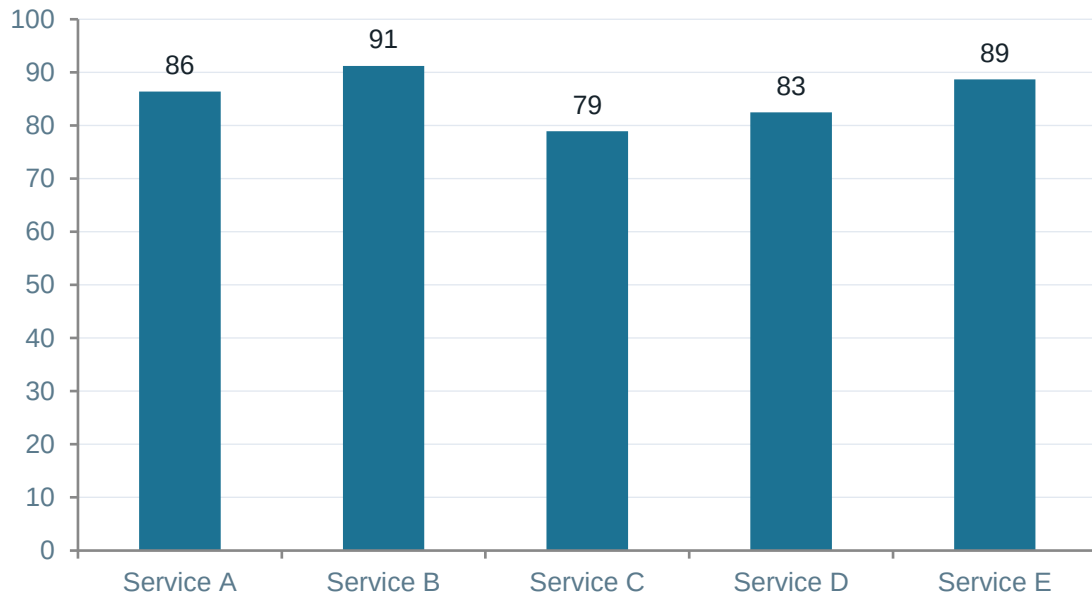
No respondent ever fills in more than one rating block.



RESULTS

Same 4-line pipeline. Any number of services.

Simulated: 15 respondents · 5 services



SCALES TO

Hundreds
services

with zero code changes — only
the input data grows

TAKEAWAY

One rating, many services.

tidyverse turns a multi-select survey response into a clean, per-service attribution pipeline — staying compliant with national regulation while respecting citizens' time.

CODE & SAMPLE DATA

github.com/Nr5D/multi-service-attribution